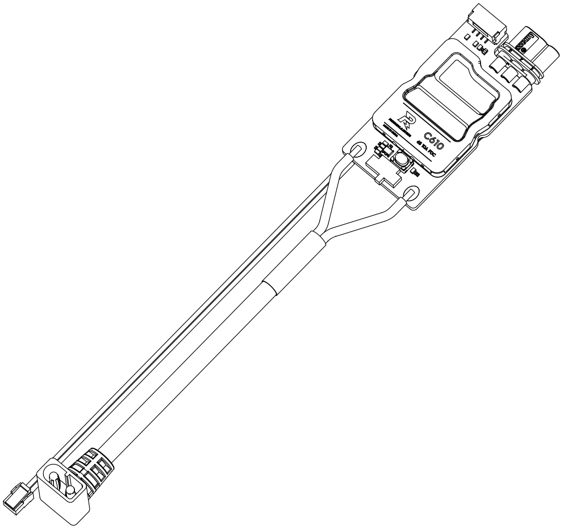
ROBOMASTER

C610 无刷电机调速器
C610 Brushless DC Motor Speed Controller

使用说明

User Guide

v1.0 2025.11



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免责声明

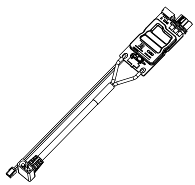
使用本产品前，请您仔细阅读本文、访问 https://www.robomaster.com 阅读本产品相关的所有指引。使用本产品视为您已经阅读并接受本文与本产品相关的全部条款。

简介

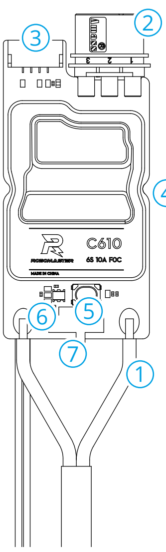
RoboMaster C610 无刷电机调速器（简称“C610 电调”）搭载 32 位定制驱动芯片，基于磁场定向控制 (FOC) 技术实现电机转矩的精准调节。C610 电调可与 RoboMaster M2006 P36 直流无刷减速电机（简称“M2006 电机”）组成高性能动力套件，并支持通过 RoboMaster Assistant 调参软件配置参数及升级固件。

物品清单

C610 电调 × 1



部件介绍



1. 电源线

   连接电源（额定电压 24V）为 C610 电调供电。

2. 三相动力接头

   与 M2006 电机的三相动力接头相连接，连接时请确保电调与电机连线正确。请勿强行反插接头。

3. 4-Pin 位置传感器数据端口

   连接 M2006 电机 4-Pin 位置传感器数据线，以获取位置传感器数据。

4. 电调固定槽

   电调对称位置共两个固定槽。可使用 M2 螺丝通过该固定槽固定电调。

5. SET 按键

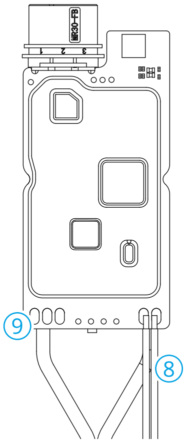
   对电调进行配置，详见 SET 按键操作。

6. 指示灯

   指示当前电调的工作状态，详见指示灯描述。

7. CAN 终端电阻选择开关

   通过拨动开关至 ON/NC 位置，可接入 / 断开 120Ω 终端电阻（用户可参阅 CAN 总线布线和终端电阻选择的相关规范，选择终端电阻是否接通）。



8. CAN 信号线

从左到右依次为：黑色 (CAN_L) 和红色 (CAN_H)。

将 CAN 信号线连接到控制板接收 CAN 控制指令，CAN 总线比特率为 1Mbps。

9. **串口调参端口**

从左到右依次为：GND、TXD 和 RXD。

通过焊接导线或其它方式将 USB 转串口模块连接到电调，再将 USB 转串口模块连接到 PC，使用 RoboMaster Assistant 调参软件对电调进行固件升级及参数设置。

⚠ • 在使用 CAN 总线指令控制时，用户可以通过 CAN 总线获取转子的位置、转速等关键参数。在使用 CAN 总线时，请考虑总线带宽，合理使用总线资源。

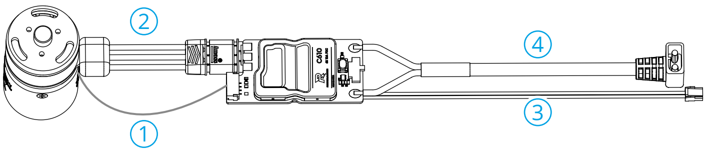
产品使用

注意事项

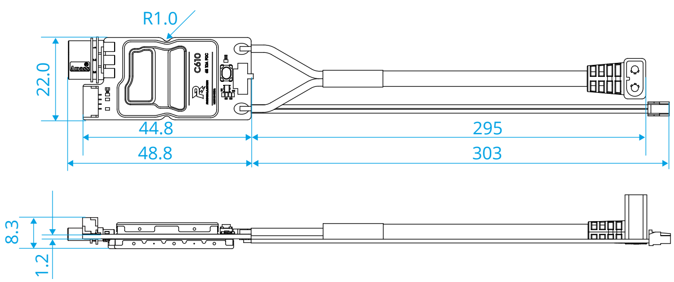
⚠
- 确保电路无短路、接口按照要求正确连接。
- 电调大功率输出时，会出现发热的情况，请小心使用，避免烫伤。
- 本产品不具备反电动势 / 反向电流吸收能力，刹车瞬间会使电源线电压升高，甚至损坏本产品。建议在电调输入端设置反电动势 / 反向电流吸收模块，确保将电源线电压钳制在 35V 以内。
- 在电调电源线接通额定电压 24V 前，禁止通过其他端口给电调供电。
- 请严格按照本文规定的工作环境（如电压、电流、温度等参数）使用，否则将会影响产品寿命或造成永久性损坏。

连接

1. 将电机的 4-Pin 位置传感器数据线插入电调的 4-Pin 位置传感器数据端口。
2. 将电机的三相动力线与电调三相动力接头相连接，连接时请确保电调与电机连线正确。请勿强行反插接头。
3. 将 CAN 信号线连接到控制板的 CAN 信号接口。
4. 连接电源线至电源为电调进行供电。



安装



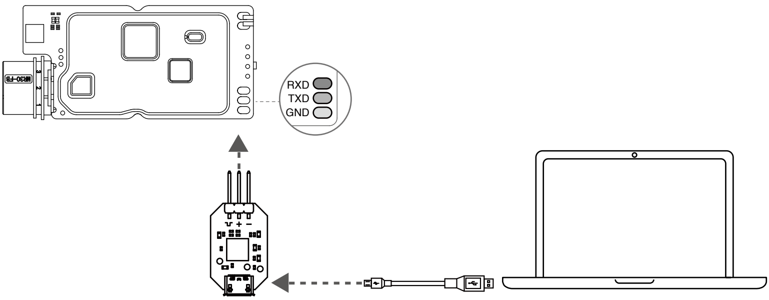
单位：mm

电调侧面对称放置 2 个半圆形凹槽，用户可用该凹槽固定电调，内槽孔直径为 2mm。若需要固定，建议使用 M2 螺丝。

使用调参软件

使用 USB 转串口工具（例如 DJI Takyon 电调升级器），连接电调至计算机，以使用 RoboMaster Assistant 对电调进行参数设置或固件升级。下图以使用 Takyon 电调升级器为例进行说明。

⚠ • USB 转串口的电平必须为 3.3 V。



1. 从 RoboMaster 官网下载 RoboMaster Assistant 调参软件。
2. 将 USB 转串口模块按照线序连接到 C610 电调上，电调 GND、TXD、RXD 分别与 Takyon 电调升级器的“-”、“+”、和“ ⎍ ”相联。然后连接 USB 转串口模块到计算机。
3. 连接电源为电调供电，设置完成前切勿切断电源或连接。
4. 运行 RoboMaster Assistant 调参软件。软件界面显示已连接设备，表示电调与软件已连接并能正常通信。
5. 使用 RoboMaster Assistant 调参软件的设置界面进行基本参数的设置。
6. 点击固件升级按钮进行相应的固件版本升级，RoboMaster Assistant 调参软件将自行下载并升级固件。

指示灯描述

请根据电调指示灯状态判断电调工作情况。当警告及异常情况同时出现时，电调指示灯仅指示异常状态。若同时存在多个警告或异常状态，电调状态指示灯将按照闪烁次数最少的状态进行提示。在异常状态下，电调将关闭输出。

正常状态	描述
绿灯每隔 1 秒闪 N 次	当前电调的 ID 为 N，电调 ID 范围为 1 到 8
快速设置 ID 状态	**描述**
橙灯常亮	当前电调处于快速设置 ID 状态
电机校准状态	**描述**
绿灯快闪	当前电调处于校准模式
警告状态	**描述**
橙灯每隔 1 秒闪 2 次	总线上有相同 ID 的设备
异常状态	**描述**
红灯每隔 1 秒闪 1 次	电调供电电压过高（仅开机自检一次）
红灯每隔 1 秒闪 2 次	电机三相线未接入
红灯每隔 1 秒闪 3 次	与电机相连的 4-Pin 位置传感器数据线信号丢失
红灯每隔 1 秒闪 4 次	电机堵转
红灯快闪	电机校准失败

SET 按键操作

1. 独立设置 ID

用户对单个 C610 电调进行 ID（支持范围 1-8）设置，具体操作如下：

a. 电调正常工作状态下，短按 1 次 SET 按键，进入独立设置 ID 模式，此时指示灯熄灭。

b. 在独立设置 ID 模式下，短按 SET 按键的次数（不超过 8 次）即为设置的 ID 号。每次有效短按，指示灯橙灯闪烁 1 次。

c. 若 3 秒未对 SET 按键进行操作，电调将自动保存当前设置 ID 号。设置完 ID 的电调需要重新上电才能进入正常工作状态。

⚠ • 同一总线上不能出现 ID 重复的情况，否则 ID 冲突的电调将提示警告、关闭输出。

2. 快速设置 ID

用户对总线上的所有 C610 电调（不超过 8 个）进行快速编号，具体操作如下：

a. 正常工作状态下，对总线上任意 1 个 C610 电调的 SET 按键进行 1 次短按，进入独立 ID 设置模式后，再长按 SET 按键，此时总线上的所有电调将进入快速设置 ID 模式，所有电调指示灯为橙灯常亮。

b. 按照预设 ID 依次手动转动 C610 电调对应的 M2006 电机的转子（任意方向至少旋转半圈以上），电调会按照转动顺序自动从 1 依次开始编号，编号完成的电调需重新上电才能进入正常工作状态。

⚠ • 该模式下未设置 ID 的电机（未转动转子）重新上电后会保持原有 ID。若同一总线上出现相同 ID 的设备，电调将不会工作。

• 请注意按照 CAN 总线布线和终端电阻选择的规范正确选择接入或断开终端电阻，以免 CAN 总线通信无法正常工作，导致以上功能无法正常使用。

3. 电机校准

电调电机连接并接通电源后，用户通过对电机的位置传感器参数进行校准，以保证电机能够正常工作，具体操作如下：

a. 长按 SET 按键，直至指示灯变为绿灯快闪，释放 SET 按键。

b. 电机进入自动校准模式，待校准完成后自动退出校准模式。

⚠ • 初次使用及更换电机或电调后，请务必运行电机校准程序。电机校准时会转动，切勿触碰，建议该操作在空载下进行。若多次校准失败，请更换电机。

CAN 通信协议

1. 电调接收报文格式

用于向电调发送控制指令控制电调的电流输出，两个标识符（0x200 和 0x1FF）各自对应控制 4 个 ID 的电调。控制转矩电流值范围 -10000~0~10000，对应电调输出的转矩电流范围 -10~0~10A。

标识符：0x200

帧格式：DATA

帧类型：标准帧

DLC：8 字节

数据域	内容	电调 ID
DATA[0]	控制电流值高 8 位	1
DATA[1]	控制电流值低 8 位	
DATA[2]	控制电流值高 8 位	2
DATA[3]	控制电流值低 8 位	
DATA[4]	控制电流值高 8 位	3
DATA[5]	控制电流值低 8 位	
DATA[6]	控制电流值高 8 位	4
DATA[7]	控制电流值低 8 位	

标识符：0x1FF

帧格式：DATA

帧类型：标准帧

DLC：8 字节

数据域	内容	电调 ID
DATA[0]	控制电流值高 8 位	5
DATA[1]	控制电流值低 8 位	
DATA[2]	控制电流值高 8 位	6
DATA[3]	控制电流值低 8 位	
DATA[4]	控制电流值高 8 位	7
DATA[5]	控制电流值低 8 位	
DATA[6]	控制电流值高 8 位	8
DATA[7]	控制电流值低 8 位	

2. 电调反馈报文格式

电调向总线上发送的反馈数据。

标识符：0x200 + 电调 ID（如：ID 为 1，该标识符为 0x201）

发送频率：1KHz （默认值，可在 RoboMaster Assistant 软件中修改发送频率）

帧格式：DATA

帧类型：标准帧

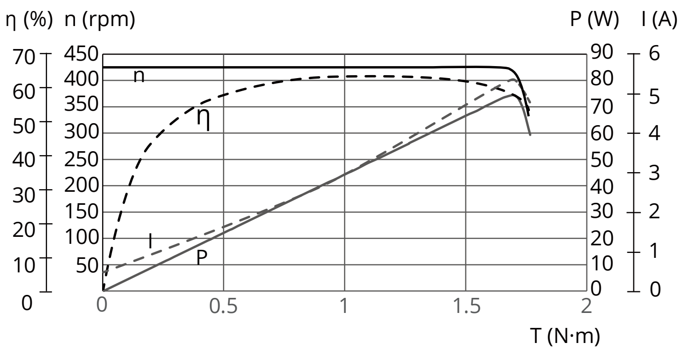
DLC：8 字节

数据域	内容
DATA[0]	转子机械角度高 8 位。角度值范围：0 ~ 8191（对应 0~360° ）
DATA[1]	转子机械角度低 8 位。角度值范围：0 ~ 8191（对应 0~360° ）

DATA[2]	转子转速高 8 位（单位：rpm）
DATA[3]	转子转速低 8 位（单位：rpm）
DATA[4]	实际转矩电流高 8 位
DATA[5]	实际转矩电流低 8 位
DATA[6]	空
DATA[7]	电机错误码： 0：无异常 2：电调供电电压过高（仅开机自检） 3：电机三相线未接入 4：与电机相连的数据线中位置传感器数据丢失 6：电机堵转 7：电机校准失败 若同时存在多个警告或异常状态，优先反馈级别更高（数值更小）的错误码。

性能与参数

搭配 M2006 电机时的电机性能曲线



n- 转速，η- 效率，I- 电流，P- 输出功率，T- 扭矩

以上数据均为输入电压 24V、在室温 25°C、通风良好的实验环境下获得，仅供参考。实际使用时，请根据工作环境温度、散热条件控制等实际情况使用。

电调参数

项目	参数
额定电压 (DC)	24 V
最大允许电流*（持续）	10 A
CAN 总线比特率	1Mbps
重量	21 g
尺寸（长 × 宽 × 高）	48.8 × 22.0× 8.3 mm
工作环境温度范围	0 至 55 ℃

* 室温 25℃、通风良好的实验环境下测得。

Disclaimer

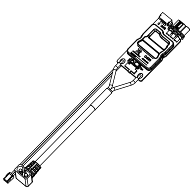
By using this product, you signify that you have read, understand, and accept the terms and conditions of this guideline and all instructions at https://www.robomaster.com. THE PRODUCT AND ALL MATERIALS AND CONTENT AVAILABLE THROUGH THE PRODUCT ARE PROVIDED "AS IS" AND ON AN "AS AVAILABLE" BASIS WITHOUT WARRANTY OR CONDITION OF ANY KIND.

Introduction

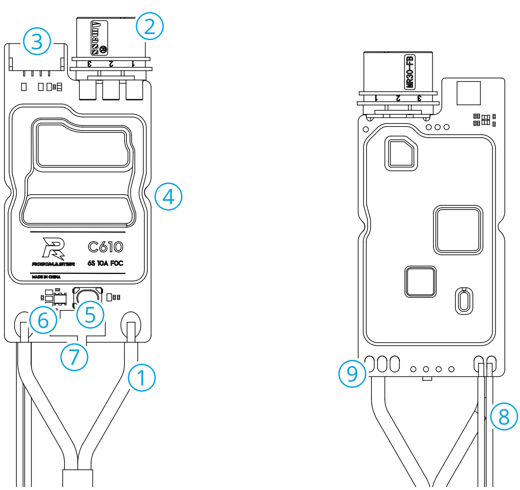
RoboMaster C610 Brushless DC Motor Speed Controller (hereinafter referred to as "C610 MSC") uses a 32-bit customized driver chip and Field-Oriented Control (FOC) to precisely control motor torque. C610 MSC and RoboMaster M2006 P36 Brushless DC Gear Motor (hereinafter referred to as "M2006 Motor") form a high-performance power unit. You can use the RoboMaster Assistant software to configure parameters and update firmware.

In the Box

C610 MSC	× 1



Overview



1. **Power Cable**

   Connect a power supply to C610 MSC (rated voltage: 24 V).

2. **Three-Phase Cable Connector**

   Connect to the three-phase cable connector of M2006 Motor, ensuring that the wires match. DO NOT insert the connectors in the wrong direction.

3. **4-Pin Position Sensor Port**

   Connect to the 4-pin position sensor cable of M2006 Motor to obtain position sensor data.

4. **Mounting Recess**

   Two mounting recesses are symmetrically distributed. Use M2 screws in the mounting recesses to secure C610 MSC.

5. **SET Button**

   Press to configure MSC parameters. For details, see Using the SET Button.

6. **Status LED**

   For details, see Status LED.

7. **CAN Terminal Resistor Switch**

   Toggle to the ON/NC position to connect/disconnect the 120Ω terminal resistor. Refer to the relevant CAN bus wiring and terminal resistor selection guidelines to determine whether the terminal resistor should be enabled.

8. **CAN Signal Wires**

   From left to right: black (CAN_L) and red (CAN_H).

   Connect the CAN signal wires to the control board to receive CAN control commands. The CAN bus bitrate is 1 Mbps.

9. **Serial Port**

   From left to right: GND, TXD, and RXD.

   Use a USB-to-serial converter to connect the MSC to a computer by soldering wires or using other suitable methods. Once connected, use the RoboMaster Assistant software to update the firmware and configure MSC parameters.

⚠ • When using CAN bus commands for control, you can also obtain key parameters such as rotor position and speed from the CAN bus. In this case, manage bus bandwidth and resources appropriately.

Using the Product

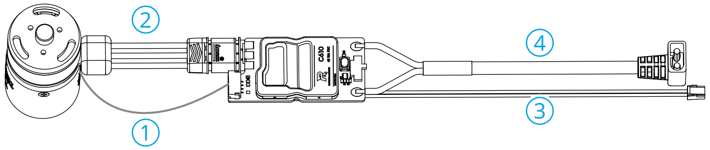
Precautions

⚠

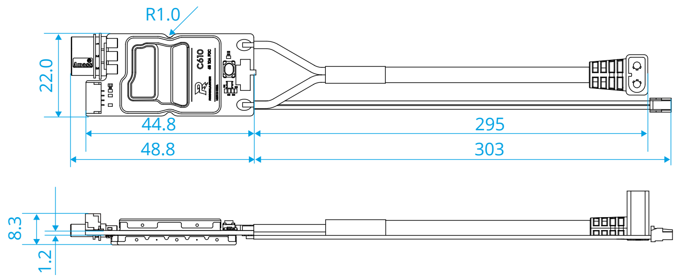
- Make sure that there are no short circuits and that all connectors are properly connected.
- The MSC may become hot during high-power operation. Avoid direct contact to prevent burns.
- The product does not have reverse-current absorption capability to handle back-EMF (electromotive force). During braking, the back-EMF generated by the motor may cause a transient voltage rise on the power cable, which can damage the product. It is recommended to place a back-EMF/reverse current absorption module at the MSC power input. This module should clamp the power cable voltage to within 35 V.
- Before connecting the power cable to a 24 V supply, DO NOT power the MSC through any other port.
- Use the product only within the specified operating conditions (including voltage, current, and temperature) to prevent reduced service life or permanent damage.

Connecting the MSC

1. Insert the 4-pin position sensor cable of the motor into the 4-pin position sensor port on the MSC.
2. Connect the three-phase cable connectors of the MSC and the motor, ensuring that the wires match. DO NOT insert the connectors in the wrong direction.
3. Connect the CAN signal wires to the CAN port on the control board.
4. Connect the power cable to the power supply.



Installation



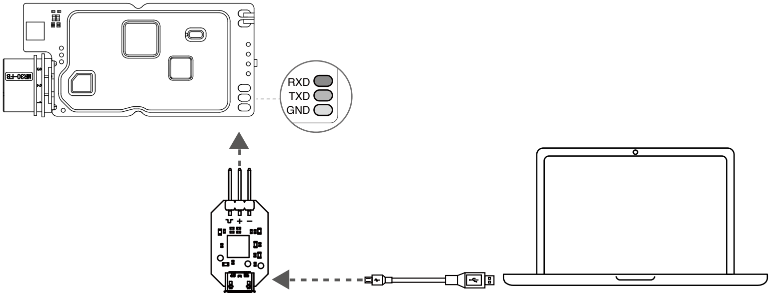
Unit: mm

The MSC has two semicircular mounting recesses symmetrically distributed on its sides. Each recess has an inner hole diameter of 2 mm. It is recommended to use M2 screws for installation.

Using RoboMaster Assistant

To configure parameters and update firmware using the RoboMaster Assistant software, connect the MSC to a computer using a USB-to-serial converter (such as the DJI Takyon Updater). The DJI Takyon Updater is used as an example in the following instructions.

⚠ • The signal-level voltage of the USB-to-serial converter must be 3.3 V.



1. Download the RoboMaster Assistant software from the official website.
2. Connect the USB-to-serial converter to C610 MSC, ensuring that the wire sequence is correct. In this example, connect the GND, TXD, and RXD pins of the MSC to the "-", "+", and " ⎍ " terminals of the Takyon Updater, respectively. Then, connect the USB-to-serial converter to the computer.
3. Connect the MSC to the power supply. DO NOT disconnect the power or any cables until all settings are complete.

4. Launch the RoboMaster Assistant software and check whether the MSC is successfully connected.
5. Use the settings page to configure basic parameters.
6. Click the firmware update button. The software will automatically download and update the MSC to the selected firmware version.

Status LED

The status LED indicates the operating state of the MSC. If a warning and an error occur at the same time, the LED indicates only the error. If multiple warnings or errors occur at the same time, the LED indicates the state with the lowest blink count. When an error occurs, the MSC disables output to the motor.

Normal	**Description**
Blinking green N times per second	N is the current MSC ID. Valid ID range: 1 to 8
Quick ID Setup	**Description**
Solid orange	The MSC is in Quick ID Setup mode
Motor Calibration	**Description**
Blinking green quickly	The MSC is in calibration mode
Warning	**Description**
Blinking orange twice per second	The current MSC ID already exists on the bus
Error	**Description**
Blinking red once per second	The MSC supply voltage is too high (detected only once during power-on self-test)
Blinking red twice per second	The three-phase cable connector is not connected to the motor
Blinking red three times per second	The signal is lost on the 4-pin position sensor cable connected to the motor
Blinking red four times per second	The motor is stalled
Blinking red quickly	Motor calibration failed

Using the SET Button

1. Standalone ID Setup

Sets the ID for a single C610 MSC. Valid ID range: 1 to 8.

a. When the MSC is operating normally, press the SET button once to enter Standalone ID Setup mode. The status LED turns off.

b. Press the SET button the same number of times as the desired ID (maximum 8). The status LED blinks orange once after each valid press.

c. If no action is taken on the SET button for 3 seconds, the MSC automatically saves the current ID and exits Standalone ID Setup mode. After setting the ID, restart the MSC to resume normal operation.

⚠ • Duplicate IDs are not allowed on the same bus. Otherwise, the MSC will report an error and disable the output to the motor.

2. Quick ID Setup

Quickly assigns IDs to all C610 MSCs (up to 8) on the bus.

a. When the MSC is operating normally, press the SET button on any C610 MSC once to enter Standalone ID Setup mode, then press and hold the SET button to enter Quick ID Setup mode. The status LED turns solid orange.

b. Manually rotate the rotor of the corresponding M2006 motor connected to each ESC (in any direction, at least half a revolution). IDs are assigned in the order in which the rotors are rotated, starting from 1. After ID assignment, restart each MSC to resume normal operation.

⚠ • In this mode, any MSC whose rotor is not rotated (no ID assigned) keeps its original ID after restart. MSCs with duplicate IDs on the same bus do not operate.

• Follow the CAN bus wiring and terminal resistor selection guidelines to ensure reliable communication.

3. Motor Calibration

Calibrates the motor position sensor parameters after connecting the MSC to the motor and powering on.

a. Press and hold the SET button until the status LED blinks green quickly, then release the button.

b. The motor enters automatic calibration mode and exits the mode once calibration is complete.

⚠ • Perform motor calibration at first use or after replacing the motor or MSC. The motor rotates during calibration. DO NOT touch the motor during this process. It is recommended to perform calibration with no load attached. If calibration fails multiple times, replace the motor.

CAN Communication Protocol

1. Receiving Message Format

Use this format to send control commands to the MSC to adjust its output current. Two identifiers (0x200 and 0x1FF) are used, and each identifier can control four MSC IDs. The value range of the commanded quadrature current is -10000~0~10000, and the corresponding value range of the output quadrature current is -10~0~10 (unit: A).

Identifier: 0x200

Frame format: DATA

Frame type: Standard frame

DLC: 8 bytes

Data Field	Content	MSC ID
DATA[0]	Upper 8 bits of the commanded current value	1
DATA[1]	Lower 8 bits of the commanded current value	
DATA[2]	Upper 8 bits of the commanded current value	2
DATA[3]	Lower 8 bits of the commanded current value	
DATA[4]	Upper 8 bits of the commanded current value	3
DATA[5]	Lower 8 bits of the commanded current value	
DATA[6]	Upper 8 bits of the commanded current value	4
DATA[7]	Lower 8 bits of the commanded current value	

Identifier: 0x1FF

Frame format: DATA

Frame type: Standard frame

DLC: 8 bytes

Data field	Parameter	MSC ID
DATA[0]	Upper 8 bits of the commanded current value	5
DATA[1]	Lower 8 bits of the commanded current value	
DATA[2]	Upper 8 bits of the commanded current value	6
DATA[3]	Lower 8 bits of the commanded current value	
DATA[4]	Upper 8 bits of the commanded current value	7
DATA[5]	Lower 8 bits of the commanded current value	
DATA[6]	Upper 8 bits of the commanded current value	8
DATA[7]	Lower 8 bits of the commanded current value	

2. Sending Message Format

Use this format to send data from the MSC to the CAN bus.

Identifier: 0x200 + MSC ID (for example, if MSC ID is 1, the identifier is 0x201)

Transmission frequency: 1 KHz (default, configurable in RoboMaster Assistant)

Frame format: DATA

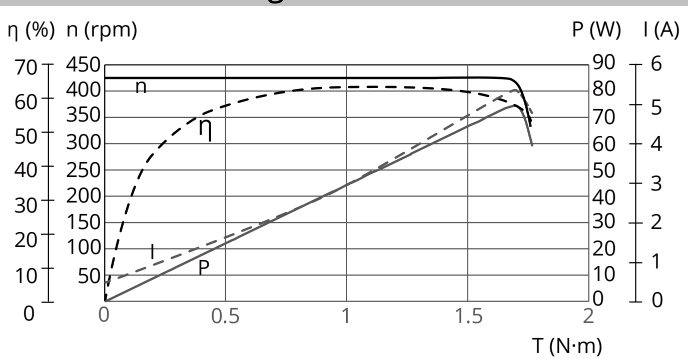
Frame type: Standard frame

DLC: 8 bytes

Data field	Parameter
DATA[0]	Upper 8 bits of the rotor mechanical angle. Angle range: 0 to 8191 (corresponding to 0° to 360°)
DATA[1]	Lower 8 bits of the rotor mechanical angle. Angle range: 0 to 8191 (corresponding to 0° to 360°)
DATA[2]	Upper 8 bits of the rotor speed (unit: rpm)
DATA[3]	Lower 8 bits of the rotor speed (unit: rpm)
DATA[4]	Upper 8 bits of the actual quadrature current
DATA[5]	Lower 8 bits of the actual quadrature current
DATA[6]	Reserved
DATA[7]	Motor error codes: 0: No abnormality 2: The MSC supply voltage is too high (detected only once during power-on self-test) 3: The three-phase cable connector is not connected to the motor 4: The signal is lost on the 4-pin position sensor cable connected to the motor 6: The motor is stalled 7: Motor calibration failed If multiple warnings or errors occur at the same time, the error code with higher priority (smaller value) will be reported first.

Performance and Parameters

Performance When Using M2006 Motor



n-Rotational Speed, η-Efficiency, I-Current, P-Output Power, T-Torque

The above data were measured under laboratory conditions at an input voltage of 24 V, an ambient temperature of 25° C (77 °F), and under normal cooling. They are provided for reference only. Adjust the operating time based on the actual conditions such as operating temperature and cooling conditions.

MSC Parameters

Item	Value
Rated Voltage (DC)	24 V
Maximum Input Current * (Continuous)	10 A
CAN Bus Bitrate	1 Mbps
Weight	21 g
Dimensions (L×W×H)	48.8 × 22.0 × 8.3 mm
Operating Temperature	0° to 55° C (32° to 131° F)

* Measured under laboratory conditions at an ambient temperature of 25 °C (77 °F) and under normal cooling.